POLYMERS 1

1

The word **polymer** comes from the Greek, meaning **having many parts**. Polymers are large molecules made up of many repeating subunits called **monomers** (*poly*: many; *mono*: one). Consider a metal chain. Imagine a polymer as representing the entire chain and each link as representing one monomer.

Definitions of a polymer

- A molecule consisting of repeated subunits
- A large molecule formed by the combination of many similar, smaller molecules
- A molecule of high relative molecular mass, made up of repeating molecules of low relative molecular mass.

1 Give another definition of a *polymer*.

Natural polymers

When people refer to polymers, they often mean artificial polymers (also referred to as 'plastics'). There are many natural polymers; common ones include starch, cellulose, proteins, wool, silk, cotton and rubber.

Starch is made up of thousands of repeating glucose units. Glucose is a sugar (saccharide) and its structure is shown in figure 1. Starch is made up of glucose units (slightly altered) and has the formula $(C_6H_{10}O_5)_n$.

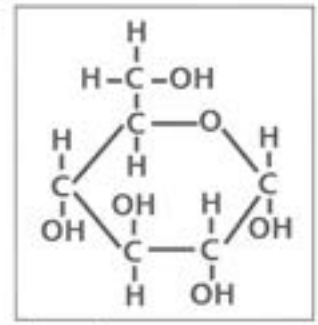


Figure 1

2 Copy figure 1. Beside your diagram write the molecular formula of glucose.

3 Starch is a polysaccharide. What does *polysaccharide* mean?

4 In the formula for starch, $(C_6H_{10}O_5)_n$, what does the *n* stand for?

Many plants store starch as a food source and this energy source can be utilised by humans. **Cellulose** is another polymer found in plants. It is a long chain polymer with glucose molecules as the monomer, but cellulose is not digestible by humans. The glucose monomers in this polymer are joined in a different way to those in starch. Cellulose has the formula $(C_6H_{10}O_5)_n$ and it is the common material of plant cell walls. It provides the structural strength for natural fibres; e.g. cotton is 95% cellulose.

5 Name three plants rich in starch.

6 Give two similarities and two differences between starch and cellulose.

Proteins are polymers made up of amino acid monomers. Silk is a natural protein fibre obtained from the cocoons produced by silkworms. Traditionally, plant (and animal fibres) have been spun and woven into cloth.

7 Give one use for silk material.

8 Name one animal fibre used in clothing.

Natural rubber is obtained from **latex**, a milky emulsion that occurs in the sap of certain tropical plants. The major source of natural rubber is the para rubber tree. When an incision is made into the bark of the tree, latex oozes out. The tree responds to the wound by producing more latex. The latex is treated to form solid rubber and then processed.

9 Give two uses for rubber.

Natural rubber has drawbacks, as it is especially sensitive to temperature. It becomes soft when warm and brittle when cold. It also perishes (loses structure and strength) over time. The addition of sulfur to rubber (vulcanisation) was patented by Charles Goodyear in the middle of the nineteenth century. This process stabilises and improves the properties of rubber by cross-linking the individual polymer molecules. Isoprene (2-methyl-1, 3-butadiene) is the major monomer in natural rubber. Its structural formula is shown in figure 2.

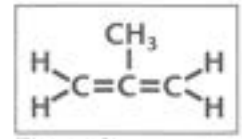


Figure 2

10 Give the molecular formula of isoprene.

11 Give three disadvantages of natural rubber.

12 What is *vulcanisation*?

13 Which monomer is present in both starch and cellulose?

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Synthetic polymers

Manufacturers have always been interested in finding alternative materials to limited resources or substances that are expensive to produce. One example is ivory – the tusks (enlarged teeth) of elephants. Ivory is a hard, white, opaque substance.

14 What is meant by *opaque*?

Ivory has been used for billiard balls, ornaments and (white) piano keys. It is a finite resource and many people have concerns about killing animals in order to obtain a decorative substance. In the mid nineteenth century, Alexander Parkes was one of the first people to develop an alternative to ivory. He invented **pyroxylin** which is also known as **Parkesine**. In Parkes's process, cellulose is dissolved and then treated with nitric acid.

15 Name one source of cellulose.

James Wesley Hyatt used cellulose to make **celluloid**. Remember, '-oid' means 'like', therefore celluloid is cellulose-like. The term celluloid refers to a class of compounds made from cellulose, camphor (derived from laurel trees) and nitric acid. One of the first uses of celluloid was making false teeth (dentures). However, the celluloid tasted of camphor and the dentures tended to soften when the wearer drank hot liquids. Celluloid became a replacement for ivory and tortoiseshell. At the beginning of the twentieth century, celluloid was adapted and produced in strip form for the newly-emerging motion picture industry. However, celluloid was very flammable and also tended to decompose over time.

16 Give two uses of ivory.

17 Give two reasons for finding a substitute for ivory.

18 Give two desirable properties of a material used for photographic film.

Cellulose has also been used as the raw material for making fibres for clothing, especially in making artificial 'silks'. Silk has long been a desirable material but it is expensive. The first artificial silks had many of the properties of the real material. However, like celluloid, they were highly flammable. Eventually a safer form of artificial silk was made: rayon. Although rayon has now been superseded by better artificial fibres, it is still in production because it is cheap and can be blended with both natural fibres and other synthetic fibres.

Research **19** What is the source of silk? How is silk thread obtained from this source?

20 Suggest a reason why fibres are blended.

Bakelite™: This artificial polymer is considered the first true **'plastic'** because it is not based on any molecules already present in nature. In physics, the word 'plastic' has a different meaning to what you might expect. It is an adjective (or 'describing' word). Using the physics definition below, plastic materials would include plasticine™ and mild steel. When a force is applied to them, these materials change shape and keep the new shape.

21 Give two materials that are 'plastic' according to the physics definition.

22 Give your own definition of *plastic*.

Plastic: the property of a material that causes it to change shape when in contact with a force, and to remain that shape until another force is applied.

Leo Hendrik Baekeland invented Bakelite™ in 1909. He discovered that when phenol and methanal (formaldehyde) were mixed together and heated, a hard solid formed. This solid is a tough, heat-resistant and cheap material that was first used in electrical and mechanical parts. Similar polymers (phenolic resins) are still used to make electronic printed circuit boards.

23 Figure 3 shows the structures of phenol and methanal. Copy the diagrams and write the molecular formula of each substance.

24 Give two properties of bakelite.

In everyday use, we tend to use the term **plastic** to refer to synthetic polymers that can be moulded.

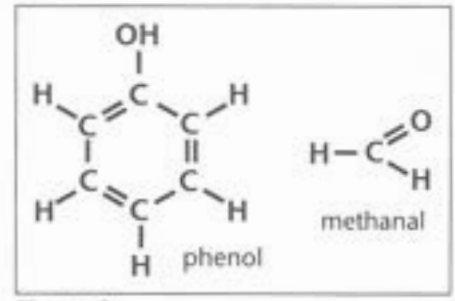


Figure 3

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